Akoizo Original Practice - Form 2

Division B | Disease Detectives | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) What strengthened Snow's work beyond a map?

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

A. Ignoring denominators

B. Assuming all deaths shared one source

C. Exposure-based population comparisons

D. Only a memorable story

Answer: _____

2. (2 points) Selecting 50 cases and 50 controls primarily defines…

A. A complete cohort automatically

B. A time series only

C. A case-control design

D. A randomized trial

Answer: _____

3. (2 points) The null value for an ordinary RR is…

Unit: Interpreting results and interrupting transmission.
Worked-case data: Observed risks are 0.4 and 0.1 in a closed cohort. Explain the measures and propose the next analytic step.

A. 100

B. -1

C. 1

D. 0

Answer: _____

4. (2 points) Which must hold for ordinary causation?

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

A. The largest group must be exposed

B. The map must be colorful

C. Relevant exposure precedes outcome

D. Outcome always precedes exposure

Answer: _____

5. (2 points) A full banquet roster investigated afterward is…

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

A. An ecological study only

B. A randomized trial

C. A retrospective cohort

D. Necessarily case-control

Answer: _____

6. (2 points) 0.4 minus 0.1 is…

A. 3 percentage points

B. A 0.3% difference

C. 30 percentage points

D. 30 times the risk

Answer: _____

7. (2 points) What can age differences produce?

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

A. No possible effect on comparisons

B. A new case definition necessarily

C. Confounding in a suitable causal structure

D. Automatic proof of causation

Answer: _____

8. (2 points) Random assignment and random sampling are…

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

A. Both proof of no bias

B. Both required for every survey

C. Different procedures

D. Synonyms

Answer: _____

9. (2 points) Vaccination primarily acts on…

Unit: Interpreting results and interrupting transmission.
Worked-case data: Observed risks are 0.4 and 0.1 in a closed cohort. Explain the measures and propose the next analytic step.

A. The map projection

B. All reporting delays

C. Host susceptibility for the relevant agent

D. Every environmental reservoir

Answer: _____

10. (4 points) Why is mapping alone insufficient?

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

Answer: _____
_____
_____

11. (4 points) Why not compute risk from an ordinary case-control sample fraction?

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

Answer: _____
_____
_____

12. (4 points) Write an RR = 4 interpretation.

Answer: _____
_____
_____

13. (4 points) Explain a comparison group's purpose.

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

Answer: _____
_____
_____

14. (4 points) Name a recall-bias example.

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

Answer: _____
_____
_____

15. (4 points) Why does significance not prove cause?

Unit: Interpreting results and interrupting transmission.
Worked-case data: Observed risks are 0.4 and 0.1 in a closed cohort. Explain the measures and propose the next analytic step.

Answer: _____
_____
_____

16. (4 points) What is confirmation bias here?

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

Answer: _____
_____
_____

17. (4 points) Explain ecological fallacy.

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

Answer: _____
_____
_____

18. (4 points) Name three chain links.

Unit: Interpreting results and interrupting transmission.
Worked-case data: Observed risks are 0.4 and 0.1 in a closed cohort. Explain the measures and propose the next analytic step.

Answer: _____
_____
_____